

Introduction to Image Processing

Homework #2 (DUE: 2026.04.01)

(You can use word, powerpoint, or pdf to organize your figures. You can also describe what you observe from your result. Please attach your codes in separate files. You can also point out the key point of your codes. Reminder: Disclosure of AI usage.)

DO NOT USE built-in function *smooth*, *imfilter*, *medfilt2*, *imsharpen*.

Image Enhancement in the Spatial Domain

Image A: hurricane-Andrew.tif

Image B: one of your own images (gray-level image)

1. Do averaging filtering (box kernel) of both image A & B. You can choose a proper kernel size to get desired smoothness depending on the size and quality of image. You can use function such as *sum* or *conv2* to accomplish filtering.

(Optional 1-1: You can try to perform total convolution, circular convolution, and using replicate padding. Observe the difference of image size and value near boundary)

(Optional 1-2: Design a Gaussian filter to provide similar smoothness as box kernel. Please show the designed Gaussian filter, kernel size and σ value.)

2. Repeat problem 1 with Median filtering and comparing the results.
3. Sharpen images A & B by using the Laplacian kernel:

1	1	1
1	-8	1
1	1	1

Show the original image, Laplacian image and sharpened image.

(Optional 3-1: You can try to perform unsharp masking and high-boost filtering and compare the results.)

For image with negative values, you can use color bar to show the range of gray level.